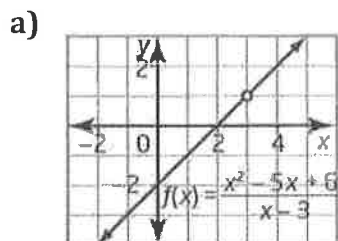


2.1/2.2 Multiplying and Dividing Rational Expressions - Worksheet

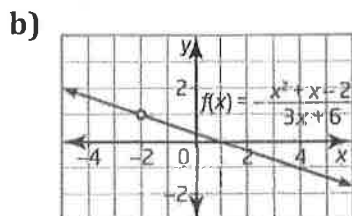
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SOLUTIONS

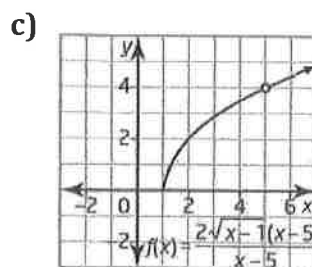
1) State the restrictions for each function.



$$x \neq 3$$

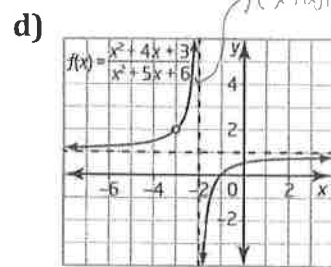


$$x \neq -2$$



$$x \neq 5$$

$$x \geq 1$$



$$x \neq -2, 3$$

2) Simplify each expression and state all restrictions on x .

a) $\frac{x-8}{x^2-13x+40}$

$$= \frac{x-8}{(x-5)(x-8)}$$

$$= \frac{1}{x-5}, x \neq 5, 8$$

b) $\frac{x^2-3x-18}{x^2+x-42}$

$$= \frac{(x-6)(x+3)}{(x+7)(x-6)}$$

$$= \frac{x+3}{x+7}, x \neq -7, 6$$

c) $\frac{x+8}{x^2+6x-16}$

$$= \frac{x+8}{(x+8)(x-2)}$$

$$= \frac{1}{x-2}, x \neq -8, 2$$

3) Simplify and state the restrictions on the variables.

a) $\frac{14y}{11x} \times \frac{121y}{7x}$

$$= \frac{22y^2}{x^2}, x \neq 0$$

b) $\frac{15b^3}{4b} \times \frac{20b}{30b^2}$

$$= \frac{5b^4}{2b^3}$$

$$= \frac{5b}{2}, b \neq 0$$

c) $\frac{5x}{9y} \div \frac{5x}{18y^2}$

$$= \frac{5x}{9y} \times \frac{18y^2}{5x}$$

$$= 2y, x \neq 0, y \neq 0$$

d) $\frac{26ab}{4a} \div \frac{39a^4b^3}{12b^4}$

$$= \frac{26ab}{4a} \times \frac{12b^4}{39a^4b^3}$$

$$= \frac{6ab^5}{3a^5b^3}$$

$$= \frac{2b^2}{a^4}, a \neq 0, b \neq 0$$

4) Simplify and state the restrictions on the variable.

a) $\frac{5x^5}{x+10} \times \frac{x+10}{5}$

$$= 5, x \neq -10$$

b) $\frac{x+5}{x-3} \times \frac{x-3}{x+7}$

$$= \frac{x+5}{x+7}, x \neq 3, -7$$

$$c) \frac{x+1}{x} \div \frac{x+1}{2x}$$

$$= \frac{\cancel{x+1}}{\cancel{x}} \cdot \frac{2\cancel{x}}{\cancel{x+1}}$$

$$= 2, x \neq 0, -1$$

$$d) \frac{x+12}{x+10} \div \frac{x+12}{x-5}$$

$$= \frac{\cancel{x+12}}{\cancel{x+10}} \cdot \frac{x-5}{\cancel{x+12}}$$

$$= \frac{x-5}{x+10}, x \neq -10, -12, 5$$

5) Simplify and state the restrictions on the variable.

$$a) \frac{3x^2}{12x^2+18x} \times \frac{4x+6}{3x+30}$$

$$= \frac{\cancel{3}x^2}{6x(\cancel{2x+3})} \cdot \frac{2(\cancel{2x+3})}{\cancel{3}(x+10)}$$

$$= \frac{\cancel{2}x^2}{3\cancel{6}x(x+10)} = \frac{x}{3(x+10)}, x \neq -10, -\frac{3}{2}, 0$$

$$b) \frac{4x+24}{x^2+8x} \times \frac{12x^2}{3x+18}$$

$$= \frac{4(\cancel{2x+6})}{x(x+8)} \cdot \frac{4\cancel{3}x^2}{\cancel{3}(x+6)}$$

$$= \frac{16x^2}{x(x+8)} = \frac{16x}{x+8}, x \neq -8, -6, 0$$

$$c) \frac{x^2+10x+21}{x+3} \times \frac{x+2}{x^2+9x+14}$$

$$= \frac{(\cancel{x+7})(\cancel{x+3})}{\cancel{x+3}} \cdot \frac{x+2}{(\cancel{x+7})(x+2)}$$

$$= 1, x \neq -3, -2, -7$$

$$d) \frac{x^2+2x-15}{x^2-9x+18} \times \frac{x-6}{x+5}$$

$$= \frac{(\cancel{x+5})(x-3)}{(\cancel{x+5})(x-3)} \cdot \frac{(x-6)}{(x+5)}$$

$$= 1, x \neq -5, 3, 6$$

6) Simplify and state the restrictions on the variable

$$a) \frac{x^2+15x}{4x+24} \div \frac{3x}{3x+18}$$

$$= \frac{x(\cancel{x+15})}{4(\cancel{x+6})} \times \frac{\cancel{3}(x+6)}{3x}$$

$$= \frac{x+15}{4}, x \neq -6, 0$$

$$b) \frac{6x}{8x-72} \div \frac{9x}{2x-18}$$

$$= \frac{\cancel{2}6x}{8(\cancel{x-9})} \cdot \frac{2(\cancel{x-9})}{3\cancel{9}x}$$

$$= \frac{4}{24}$$

$$= \frac{1}{6}, x \neq 0, 9$$

$$c) \frac{x^2+15x+26}{6x^2} \div \frac{x^2-3x-10}{30x^3}$$

$$= \frac{\cancel{(x+2)}(x+13)}{\cancel{6}x^2} \times \frac{5\cancel{3}0x^3}{(x-5)\cancel{(x+2)}}$$

$$= \frac{5x(x+13)}{x-5}, x \neq -2, 0, 5$$

$$d) \frac{x^2+11x+24}{x^2+2x-3} \div \frac{x-8}{x-1}$$

$$= \frac{(x+8)\cancel{(x+3)}}{(\cancel{x+3})(x-1)} \cdot \frac{x-1}{x-8}$$

$$= \frac{x+8}{x-8}, x \neq -3, 1, 8$$

7) Simplify and state the restrictions on the variable

$$a) \frac{a^2-25}{a+2} \cdot \frac{a^2-4}{a^2-7a+10}$$

$$= \frac{(a-5)\cancel{(a+5)}}{\cancel{a+2}} \cdot \frac{\cancel{(a-2)}(a+2)}{(\cancel{a-2})(a-5)}$$

$$= a+5, a \neq -2, 2, 5$$

$$b) \frac{y^2-4y-21}{3y^2+6y} \cdot \frac{y^2+8y}{y^2+11y+24}$$

$$= \frac{(y-7)\cancel{(y+3)}}{3y(y+2)} \cdot \frac{y(y+8)}{(\cancel{y+8})(y+3)}$$

$$= \frac{y-7}{3(y+2)}, y \neq -8, -3, -2, 0$$

$$c) \frac{p^2-2p+1}{p+1} \div \frac{p^2-1}{p+1}$$

$$= \frac{(p-1)\cancel{(p-1)}}{\cancel{p+1}} \times \frac{\cancel{p+1}}{\cancel{(p-1)}(p+1)}$$

$$= \frac{p-1}{p+1}, p \neq -1, 1$$

$$d) \frac{x^2+6x-27}{x^2+11x+18} \div \frac{x-3}{x^2+x-2}$$

$$= \frac{(x+9)\cancel{(x-3)}}{(\cancel{x+9})(x+2)} \div \frac{x-3}{(x+2)(x-1)}$$

$$= \frac{\cancel{(x+9)}\cancel{(x-3)}}{(\cancel{x+9})(x+2)} \cdot \frac{(x+2)(x-1)}{x-3}$$

$$= x-1, x \neq -9, -2, 1, 3$$

Answers

1) a) $x \neq 3$ b) $x \neq -2$ c) $x \geq 1, x \neq 5$ d) $x \neq -3, x \neq -2$

2) a) $\frac{1}{x-5}, x \neq 5, x \neq 8$ b) $\frac{x+3}{x+7}, x \neq -7, x \neq 6$ c) $\frac{1}{x-2}, x \neq -8, x \neq 2$

3) a) $\frac{22y^2}{x^2}, x \neq 0$ b) $\frac{5b}{2}, b \neq 0$ c) $2y, x \neq 0, y \neq 0$ d) $\frac{2b^2}{a^4}, a \neq 0, b \neq 0$

4) a) $5, x \neq -10$ b) $\frac{x+5}{x+7}, x \neq -7, x \neq 3$ c) $2, x \neq -1, x \neq 0$ d) $\frac{x-5}{x+10}, x \neq -12, x \neq -10, x \neq 5$

5) a) $\frac{x}{3(x+10)}, x \neq -10, -\frac{3}{2}, 0$ b) $\frac{16x}{x+8}, x \neq -8, -6, 0$ c) $1, x \neq -7, -3, -2$ d) $1, x \neq -5, 3, 6$

6) a) $\frac{x+15}{4}, x \neq -6, 0$ b) $\frac{1}{6}, x \neq 0, 9$ c) $\frac{5x(x+13)}{x-5}, x \neq -2, 0, 5$ d) $\frac{x+8}{x-8}, x \neq -3, 1, 8$

7) a) $a + 5, a \neq 2, -2, 5$ b) $\frac{y-7}{3(y+2)}, y \neq -8, -3, -2, 0$ c) $\frac{p-1}{p+1}, p \neq -1, 1$ d) $x - 1, x \neq -9, -2, 1, 3$